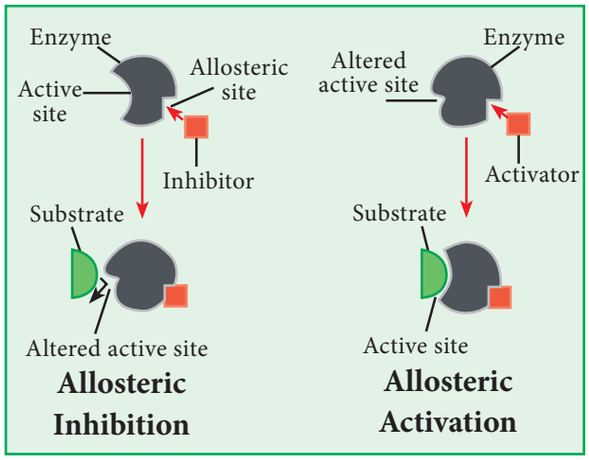


Figure 4.7 Allosteric inhibition

4.5.1 Types of Inhibition

i) Competitive Inhibition

A competitive inhibition usually is reversible. A competitive inhibitor usually closely resembles the substrate and is regarded as substrate analogue. The inhibitor competes with substrate and binds at the active site of the enzyme but does not undergo any catalysis. As long as the competitive inhibitor is bound to the active site, the enzyme will not be available for the substrate to bind. This type of inhibition can be reversed by increasing the concentration of substrate.

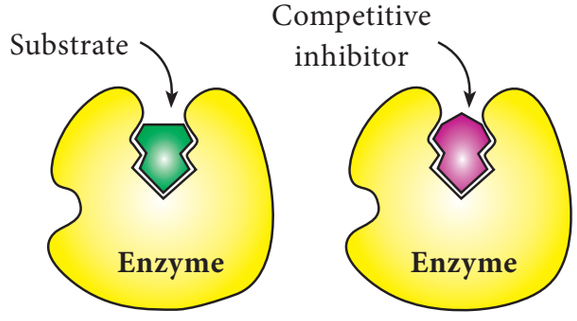


Figure 4.8 Competitive inhibition

Example: 1) Enzyme - Xanthine oxidase; Substrate - Hypoxanthine; Inhibitor – Allopurinol.

Significance of the inhibitor: Used in the control of Gout to reduce excess production of uric acid from hypoxanthine.

Example: 2) Enzyme - Succinate dehydrogenase; Substrate - Succnate; Inhibitor - Malonate

ii) Noncompetitive inhibition:

Usually a noncompetitive inhibitor binds either to free enzyme or to ES complex at a site other than the active site on the enzyme surface. This results in the change of conformation of the enzyme as well as its active site, which makes the substrate unable to bind to the enzyme effectively. This type of inhibitor has no structural resemblance with the substrate like competitive inhibitors.

Non-competitive inhibitors do not interfere with the enzyme-substrate binding. But catalysis is prevented, possibly due to the distortion of enzyme conformation.

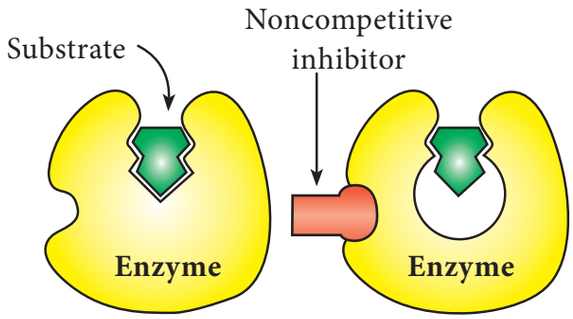


Figure 4.9 Noncompetitive inhibition

iii) Uncompetitive inhibition:

Uncompetitive inhibitors binds only to the ES complex. However, the binding of the inhibitor affects the binding of the substrate. This type of inhibition cannot be overcome. The inhibitor usually follows an allosteric effect where it binds to a different site on the enzyme than the substrate. This binding to an allosteric site changes the conformation

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